

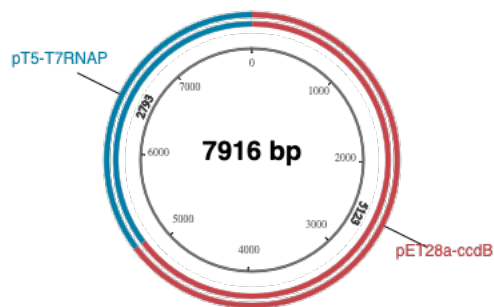
pET28a-pT5-T7RNAP

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Component Fragments

Name	Length	Produced by	5' End	3' End
pET28a-ccdB	5143	PCR	Fwd Primer (auto)	Rev Primer (auto)
pT5-T7RNAP	2813	PCR	Fwd Primer (auto)	Rev Primer (auto)



Notes

- Primer pET28a-ccdB_rev contains a run of 4+ repeats of a mono/di/trinucleotide.
- Primer pT5-T7RNAP_fwd contains a run of 4+ repeats of a mono/di/trinucleotide.
- Primer pT5-T7RNAP_fwd has %GC outside of desired range (35-65%) in the annealing segment.

Required oligos

Name	Primer 5' (overlap/spacer/ANNEAL) 3'	Len	%GC	3' %GC	3' Tm	3' Ta
pET28a-ccdB_fwd	gttcgcgtaaGCTTGATCCGGCTGCTAAC	29	55	58	66.4	65.6
pET28a-ccdB_rev	tttttatgaATTTTCGCGGGATCGAGATC	29	38	53	64.6	65.6
pT5-T7RNAP_fwd	cccgcgaaatTCATAAAAAATTTATTTGCTTTGTGAGCG	39	36	28	62.5	63.5
pT5-T7RNAP_rev	cggatcaagcTTACGCGAACGCGAAGTC	28	57	56	65.6	63.5

Build Settings

Property	Value
Product/Kit	#E5520 NEBuilder HiFi DNA Assembly Cloning Kit
Minimum Overlap	20 nt
Minimum Overlap Tm	48 °C
Circularize	Yes
PCR Polymerase/Kit	Q5 High-Fidelity DNA Polymerase
PCR Primer Conc.	500 nM
Min. Primer Length	18 nt

Assembled Sequence

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#ACCESSION   .
#VERSION     .
#KEYWORDS    NEBuilder
#SOURCE      synthetic DNA construct
# ORGANISM   synthetic DNA construct
#REFERENCE   1 (bases 1 to 7916)
#  AUTHORS   .
#  TITLE     NEBuilder-generated Construct
#  JOURNAL    Exported 10-APR-2025 from NEBuilder https://nebuilder.neb.com
#COMMENT     NEBuilder-generated oligos (UPPERCASE = gene-specific, lowercase = overlap)
#COMMENT     pET28a-ccdB_fwd: gttcgcgtaaGCTTGATCCGGCTGCTAAC
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#COMMENT     pET28a-ccdB_rev: ttttttatgaATTTCGCGGGATCGAGATC
#COMMENT     pET28a-ccdB_rev 3'Tm: 64.6 3'Ta: 65.6
#COMMENT     pT5-T7RNAP_fwd: cccgcgaaatTCATAAAAAATTTATTTGCTTTGTGAGCG
#COMMENT     pT5-T7RNAP_fwd 3'Tm: 62.5 3'Ta: 63.5
#COMMENT     pT5-T7RNAP_rev: cggatcaagcTTACGCGAACGCGAAGTC
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